Yang Runming

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Research interests: LLM post-training; Knowledge distillation; Efficient reasoning; Low-bit quantization

EDUCATION

The University of Hong Kong Oct. 2025 - Present
Ph.D. student in Electrical and Electronic Engineering — Advisor: Prof. Ngai Wong; expected Sep. 2029

Tsinghua University Sep. 2022 - Jun. 2025
M.Eng. in Big Data Engineering — Advisor: Prof. Yujiu Yang; GPA: 3.73/4.00

Huazhong University of Science and Technology Sep. 2017 - Jun. 2021
B.Eng. in AI and Automation — Science and Engineering Cross-Innovation Experimental Class

PUBLICATIONS

* Equal contribution.

[1] T. Wu*, **R. Yang***, T. Liu, J. Wang, N. Wong. *Revisiting Model Interpolation for Efficient Reasoning.* **ACL 2026 (Main)** [Paper] [arXiv] [Code]
Interpolates fast- and slow-thinking models without training to balance reasoning quality and chain-of-thought length.

[2] T. Liu*, T. Wu*, **R. Yang**, S. Sun, J. Wang, Y. Yang. *ProFit: Leveraging High-Value Signals in SFT via Probability-Guided Token Selection.* **ACL 2026 (Findings)** [Paper] [arXiv] [Code]
Uses current-policy information to guide token selection and relax token-level alignment during supervised fine-tuning.

[3] T. Wu*, Y. Cheng*, C. Ding*, **R. Yang**, X. Feng, W. Zhou, Z. Liu, N. Wong. *Can We Trust LLMs on Memristors? Diving into Reasoning Ability under Non-Ideality.* **COLM 2026** [arXiv]
Simulates how memristor non-idealities affect LLM reasoning and evaluates training-free mitigation strategies.

[4] H. Xiao, Q. Yang, D. Xie, W. Xu, Z. Su, **R. Yang**, H. Liu, W. Zhou, Z. Liu, N. Wong. *Exploring Layer-wise Information Effectiveness for Post-Training Quantization in Small Language Models.* **ACL 2026 (Findings)** [Paper] [arXiv] [Project]
Uses layer-wise information effectiveness to allocate mixed precision across layers for low-bit quantization of small language models.

[5] T. Wu, C. Tao, J. Wang, **R. Yang**, Z. Zhao, N. Wong. *Rethinking Kullback-Leibler Divergence in Knowledge Distillation for Large Language Models.* **COLING 2025 (Oral)** [Paper] [arXiv] [Code]
Compares forward and reverse KL divergence for LLM distillation and introduces Adaptive KL.

[6] **R. Yang**, T. Wu, Y. Yang. *LoCa: Logit Calibration for Knowledge Distillation.* **ECAI 2024** [arXiv]
Calibrates teacher logits for misclassified samples during knowledge distillation.

RESEARCH INTERNSHIP

Tencent AI Lab Nov. 2023 - Jun. 2025
Research Intern, Information Security Department

- Synthesized multimodal data and conducted model post-training and evaluation for image-text content moderation in QQ group chats, Qzone, and short-video comments.
- Deployed models in tiers and accelerated inference; trained models were deployed in internal risk-control systems, improving throughput and reducing false-positive rates over the previous solution.
- Researched large/small model collaboration, data distillation, and lightweight modeling; contributed to two academic papers (Shadow-FT and LLM-NEO) and one company patent.

ONGOING RESEARCH

CoT Data and Policy Distillation for Reasoning Model Post-Training Ongoing
The University of Hong Kong — Ph.D. Research

- Investigating how CoT data and policy distillation affect subsequent reinforcement learning, with a reproducible pipeline comparing FKL/RKL initialization and analyzing reward signals and computational cost.
- Initial results on a synthetic three-number arithmetic task with Qwen3-1.7B show 42.6% mean answer accuracy across three seeds for FKL distillation followed by RL, versus 21.0% for direct RL; generation-budget controls and mechanism analysis are ongoing.

PREPRINTS AND SUBMITTED MANUSCRIPTS

* Equal contribution.

[P1] **R. Yang***, T. Wu*, J. Li, P. Hu, Y.-C. Wu, N. Wong, Y. Yang. *Shadow-FT: Tuning Instruct Model via Training on Paired Base Model.* **Submitted to NAACL 2027** [arXiv] [Code]
Transfers fine-tuning updates from a base model to its paired instruction-tuned model.

[P2] Q. Yang, **R. Yang**, H. Xiao, W. Xu, J. Chen, H. Liu, C. Ding, R. Hu, Y.-C. Wu, N. Wong. *FluxBin: Flexible LUT-based Ultra-low-bit LLM Inference by Algorithm-Kernel Synergy.* **Submitted to NAACL 2027** [arXiv] [Code]
Co-designs ultra-low-bit post-training quantization and lookup-table CUDA kernels for efficient LLM inference.

[P3] T. Wu, **R. Yang**, T. Liu, J. Wang, Z. Xu, N. Wong. *Timber: Training-free Instruct Model Refining with Base Model via Effective Rank.* **Submitted to NAACL 2027** [arXiv] [Code]
Uses effective rank to analyze post-training weight changes and refine instruction-tuned models without additional training.

[P4] Y. Zhang, T. Wu, **R. Yang**, T. Liu, J. Wang, Y. Yang. *From Answers to Obligations: Benchmarking Financial Instruction Following with FinIF.* **Submitted to EMNLP 2027**
Studies instruction-following evaluation in financial tasks, focusing on whether model responses satisfy task requirements.

[P5] Z. Li*, Y. Su*, S. Wang, **R. Yang**, C. Xie, A. Liu, M. Li, J. Cao, Y. Xie, N. Wong, H. Yang. *Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning.* [arXiv]
Analyzes step-level reasoning errors in quantized LLMs and uses targeted preference pairs to improve 4-bit mathematical reasoning.

[P6] T. Liu*, T. Wu*, M. Zheng, X. Luo, **R. Yang**, X. Yang, J. Wang, Y. Yang. *Simple-OPD: Demystifying Warm-up for On-policy Distillation.* **Submitted to NAACL 2027** [arXiv] [Code]
Studies warm-up for on-policy distillation and initializes students with LoRA on teacher-generated chains of thought.

[P7] **R. Yang***, T. Wu*, J. Wang, P. Hu, Y.-C. Wu, N. Wong, Y. Yang. *LLM-NEO: Parameter Efficient Knowledge Distillation for Large Language Models.* [arXiv] [Code]
Combines low-rank adaptation and knowledge distillation for parameter-efficient knowledge transfer from large teachers to compact students.

[P8] H. Xiao*, **R. Yang***, Q. Yang*, W. Xu, Z. Li, Y. Su, Z. Liu, H. Yang, N. Wong. *PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models.* [arXiv] [Project]
Decomposes LLM weights into two ternary trit-planes with lookup-table kernels for efficient inference.